COMPLETION OF A CLOSED IMMERSION IS A CLOSED IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-closed-immersion-gives-closed-immersion; the proof environment immediately following it is its own proof; the context span delivers the section situation and cartesian square that 'in the situation above' refers to, and the template argument cited by the proof is delivered as its own card

Editorial note. The label lemma-closed-immersion-gives-closed-immersion names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the /T completion functor and its cartesian square (section-completion-functor setup, delivered as context)
- the template argument of lemma-map-completions-finite-type (same section): identify the fiber product over the completion via the cartesian square and transfer the property by base change
- base change preserves immersions (spaces-lemma-base-change-immersions of Spaces)

Source reference. Original lemma 0APV: sources/stacks-chow/tags.txt, source lines 13488–13488. Prerequisite pointer invoked at 6096; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0ANN: sources/stacks-chow/tags.txt, source lines 13446–13446. Prerequisite pointer invoked at 6045; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AQ9: sources/stacks-restricted/pinned-original-source.tex, source lines 6009–6009. Prerequisite pointer invoked at 6047; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02YW: sources/stacks-chow/tags.txt, source lines 3891–3891. Prerequisite pointer invoked at 6049; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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1. The completion functor

In this section we consider the following situation. First we fix a base scheme S. All rings, topological rings, schemes, algebraic spaces, and formal algebraic spaces and morphisms between these will be over S. Next, we fix an algebraic space X and a closed subset $T \subset |X|$. We denote $U \subset X$ be the open subspace with $|U| = |X| \setminus T$. Picture

$$U \to X \quad |X| = |U| \amalg T$$

In this situation, given an algebraic space X' over X, i.e., an algebraic space X' endowed with a morphism $f : X' \to X$, then we denote $T' \subset |X'|$ the inverse image of T and we let $U' \subset X'$ be the open subspace with $|U'| = |X'| \setminus T'$. Picture

$$U' = f^{-1}U \qquad \begin{array}{ccc} U' & \longrightarrow & X' \\ \downarrow & & \downarrow{\scriptstyle f} \\ U & \longrightarrow & X \end{array} \qquad \begin{array}{ccccc} |U'| & \longrightarrow & |X'| & \longleftarrow & T' \\ \downarrow & & \downarrow{\scriptstyle |f|} & & \downarrow \\ |U| & \longrightarrow & |X| & \longleftarrow & T \end{array} \qquad T' = |f|^{-1}T$$

We will relate properties of f to properties of the induced morphism

$$f_{/T} : X'_{/T'} \longrightarrow X_{/T}$$

of formal completions. As indicated in the displayed formula, we will denote this morphism $f_{/T}$. We have already seen that $f_{/T}$ is representable by algebraic spaces in Formal Spaces, Lemma 0APV. In fact, as the proof of that lemma shows, the diagram

$$\begin{array}{ccc} X'_{/T'} & \longrightarrow & X' \\ {\scriptstyle f_{/T}}\downarrow & & \downarrow{\scriptstyle f} \\ X_{/T} & \longrightarrow & X \end{array}$$

is cartesian. Please keep this fact in mind whilst reading the lemmas stated and proved below.

Lemma 1.1. *In the situation above. If f is a closed immersion, then $f_{/T}$ is a closed immersion.*

Proof. (Closed immersions of formal algebraic spaces are discussed in Formal Spaces, Section 0ANN.) By the same argument as in the proof of Lemma 0AQ9 this follows from Spaces, Lemma 02YW. □